

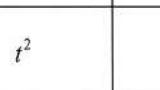
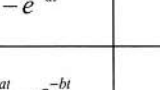
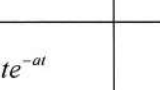
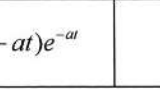
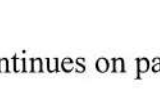
Appendix A

Properties and Table of Z Transform

使用 错误

Discrete function	z Transform
$x(k+4)$	$z^4 X(z) - z^4 x(0) - z^3 x(1) - z^2 x(2) - zx(3)$
$x(k+3)$	$z^3 X(z) - z^3 x(0) - z^2 x(1) - zx(2)$
$x(k+2)$	$z^2 X(z) - z^2 x(0) - zx(1)$
$x(k+1)$	$zX(z) - zx(0)$
$x(k)$	$X(z)$
$x(k-1)$	$z^{-1} X(z)$
$x(k-2)$	$z^{-2} X(z)$
$x(k-3)$	$z^{-3} X(z)$
$x(k-4)$	$z^{-4} X(z)$

1
1
1

	$X(s)$	$x(t)$	$x(kT)$ or $x(k)$	$X(z)$	
1.	—		Kronecker delta $\delta_0(k)$ 1, $k=0$ 0, $k \neq 0$ unit impulse function	1	3
2.	—		$\delta_0(n-k)$ 1, $n=k$ 0, $n \neq k$	z^{-k}	
3.	$\frac{1}{s}$		unit step input 单位阶跃输入 step reference signal 阶跃参考信号 $1(k)$	$\frac{1}{1-z^{-1}}$	9
4.	$\frac{1}{s+a}$		e^{-akT}	$\frac{1}{1-e^{-aT}z^{-1}}$	6
5.	$\frac{1}{s^2}$		unit ramp input 单位斜坡输入 kT	$\frac{Tz^{-1}}{(1-z^{-1})^2}$	4
6.	$\frac{2}{s^3}$		$(kT)^2$	$\frac{T^2 z^{-1}(1+z^{-1})}{(1-z^{-1})^3}$	1
7.	$\frac{6}{s^4}$		$(kT)^3$	$\frac{T^3 z^{-1}(1+4z^{-1}+z^{-2})}{(1-z^{-1})^4}$	
8.	$\frac{a}{s(s+a)}$		$1 - e^{-akT}$	$\frac{(1-e^{-aT})z^{-1}}{(1-z^{-1})(1-e^{-aT}z^{-1})}$	3
9.	$\frac{b-a}{(s+a)(s+b)}$		$e^{-akT} - e^{-bkT}$	$\frac{(e^{-aT} - e^{-bT})z^{-1}}{(1-e^{-aT}z^{-1})(1-e^{-bT}z^{-1})}$	3
10.	$\frac{1}{(s+a)^2}$		kTe^{-akT}	$\frac{Te^{-aT}z^{-1}}{(1-e^{-aT}z^{-1})^2}$	
11.	$\frac{s}{(s+a)^2}$		$(1-akT)e^{-akT}$	$\frac{1-(1+aT)e^{-aT}z^{-1}}{(1-e^{-aT}z^{-1})^2}$	

1

1

1

Note: Transform Table continues on page 8.

	$X(s)$	$x(t)$	$x(kT)$ or $x(k)$	$X(z)$	
12.	$\frac{2}{(s+a)^3}$	$t^2 e^{-at}$	$(kT)^2 e^{-akT}$	$\frac{T^2 e^{-aT} (1 + e^{-aT} z^{-1}) z^{-1}}{(1 - e^{-aT} z^{-1})^3}$	
13.	$\frac{a^2}{s^2(s+a)}$	$at - 1 + e^{-at}$	$akT - 1 + e^{-akT}$	$\frac{[(aT - 1 + e^{-aT}) + (1 - e^{-aT} - aT e^{-aT}) z^{-1}] z^{-1}}{(1 - z^{-1})^2 (1 - e^{-aT} z^{-1})}$	2
14.	$\frac{\omega}{s^2 + \omega^2}$	$\sin \omega t$	$\sin \omega kT$	$\frac{z^{-1} \sin \omega T}{1 - 2z^{-1} \cos \omega T + z^{-2}}$	6
15.	$\frac{s}{s^2 + \omega^2}$	$\cos \omega t$	$\cos \omega kT$	$\frac{1 - z^{-1} \cos \omega T}{1 - 2z^{-1} \cos \omega T + z^{-2}}$	7
16.	$\frac{\omega}{(s+a)^2 + \omega^2}$	$e^{-at} \sin \omega t$	$e^{-akT} \sin \omega kT$	$\frac{e^{-aT} z^{-1} \sin \omega T}{1 - 2e^{-aT} z^{-1} \cos \omega T + e^{-2aT} z^{-2}}$	1
17.	$\frac{s+a}{(s+a)^2 + \omega^2}$	$e^{-at} \cos \omega t$	$e^{-akT} \cos \omega kT$	$\frac{1 - e^{-aT} z^{-1} \cos \omega T}{1 - 2e^{-aT} z^{-1} \cos \omega T + e^{-2aT} z^{-2}}$	1
18.			a^k	$\frac{1}{1 - az^{-1}}$	6
19.			a^{k-1} $k = 1, 2, 3, \dots$	$\frac{z^{-1}}{1 - az^{-1}}$	2
20.			ka^{k-1}	$\frac{z^{-1}}{(1 - az^{-1})^2}$	1
21.			$k^2 a^{k-1}$	$\frac{z^{-1}(1 + az^{-1})}{(1 - az^{-1})^3}$	
22.			$k^3 a^{k-1}$	$\frac{z^{-1}(1 + 4az^{-1} + a^2 z^{-2})}{(1 - az^{-1})^4}$	
23.			$k^4 a^{k-1}$	$\frac{z^{-1}(1 + 11az^{-1} + 11a^2 z^{-2} + a^3 z^{-3})}{(1 - az^{-1})^5}$	
24.			$a^k \cos k\pi$	$\frac{1}{1 + az^{-1}}$	2
25.			$\frac{k(k-1)}{2!}$	$\frac{z^{-2}}{(1 - z^{-1})^3}$	
26.			$\frac{k(k-1)\cdots(k-m+2)}{(m-1)!}$	$\frac{z^{-m+1}}{(1 - z^{-1})^m}$	
27.			$\frac{k(k-1)}{2!} a^{k-2}$	$\frac{z^{-2}}{(1 - az^{-1})^3}$	
28.			$\frac{k(k-1)\cdots(k-m+2)}{(m-1)!} a^{k-m+1}$	$\frac{z^{-m+1}}{(1 - az^{-1})^m}$	

$x(t) = 0$, for $t < 0$.

$x(kT) = x(k) = 0$, for $k < 0$.

Unless otherwise noted, $k = 0, 1, 2, 3, \dots$

END OF PAPER

14和15

凑系数1

优先18

与18一样
优先18

2

2

2

连续时间信号定义

2

2

Continuous-Time Signal: defined over a continuous range of time, with **amplitude** of continuous range of values or discrete values.

连续时间信号:定义在一个连续的时间范围内,具有连续范围的振幅值或离散值。

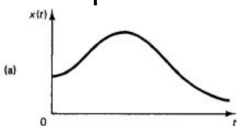
允许有离散值

模拟信号定义

Analog Signal: defined over a continuous range of time with amplitude assuming a continuous range of values. That is, continuous in both time and amplitude.

模拟信号:定义在一个连续的时间范围内,其振幅假定一个连续的值范围。也就是说,在时间和振幅上都是连续的。

Fig 1.3(a) Continuous-Time Analog Signal



量化定义

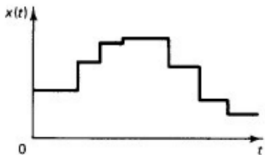
Quantization: the process of representing a variable by a set of distinct values, to obtain quantized values of the signal. Signals to be processed with a digital computer need to be quantized.

量化:用一组不同的值来表示一个变量,

从而得到信号的量化值的过程。

要用数字计算机处理的信号需要量化。

Fig 1.3(b) Continuous-Time Quantized Signal



离散信号定义

Discrete-Time Signal: defined only at discrete instants of time (i.e., quantized time). Amplitude can be continuous and discrete (quantized).

离散时间信号:只定义在离散时刻(即量化时间)。

振幅可以是连续的,也可以是离散的(量化的)。

z变换定义 2 2

$$X(z) = \mathcal{Z}[x(t)] = \sum_{k=0}^{\infty} x(kT)z^{-k}$$

部分分式展开(PFE)方法：多个极点

Z变换定理 2 2

$$\frac{X(z)}{z} = \frac{c_1}{(z-p)^2} + \frac{c_2}{z-p}$$

分母二次方
请一定拆为两项

1. 乘以一个常数

$$\mathcal{Z}[ax(t)] = a\mathcal{Z}[x(t)] = aX(z)$$

2. Z变换的线性

$$\mathcal{Z}[\alpha f(t) + \beta g(t)] = \alpha \mathcal{Z}[f(t)] + \beta \mathcal{Z}[g(t)]$$

3. 乘以a^k

$$\mathcal{Z}[a^k x(k)] = X(a^{-1}z)$$

4. 平移定理

$$\mathcal{Z}[x(t - nT)] = z^{-n}X(z)$$

5. 复平移定理 1

$$\mathcal{Z}[x(t + nT)] = z^n \left[X(z) - \sum_{k=0}^{n-1} x(kT)z^{-k} \right]$$

$$\mathcal{Z}[e^{-at}x(t)] = X(ze^{aT})$$

6. 初值定理 1 1

$$x(0) = x(t) \big|_{t=0} = \lim_{z \rightarrow \infty} X(z)$$

前提条件：要求极限存在

7. 终值定理 9

Final Value Theorem

Assume that **all poles of $X(z)$** lie *inside the unit circle*, with the possible exception of a simple pole at $z = 1$

- 1. 算Z平面的，幅度和角度
- 2. 通过Z与S的公式，将幅度和角度代入计算
- 3. 阻尼比 = -实部 / 模长

$$\lim_{k \rightarrow \infty} x(k) = \lim_{z \rightarrow 1} (1 - z^{-1})X(z)$$

Given $z = 0.4 \pm j0.4$, find the equivalent s -plane poles:

1. Compute the magnitude and angle:
 $|z| = \sqrt{0.4^2 + 0.4^2} = 0.5657, \quad \theta = \arctan\left(\frac{0.4}{0.4}\right) = 45^\circ$
2. Convert to s -domain using $s = \frac{\ln(z)}{T}$:
 $s = \frac{\ln(0.5657) + j\frac{\pi}{4}}{0.1} = -5.68 \pm j7.854$
3. Compute the damping ratio:
 $\zeta = \frac{-\operatorname{Re}(s)}{\sqrt{\operatorname{Re}(s)^2 + \operatorname{Im}(s)^2}} = \frac{5.68}{9.688} \approx 0.586$

超调量

$$e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}}} \times 100\%$$

稳态时间

$$t_s = \frac{3}{\xi \omega_n} \quad t_s = \frac{4}{\xi \omega_n} \quad t_s = \frac{4.6}{\xi \omega_n}$$

5% 2% 1%

稳态跟踪误差est
steady state tracking error
R-C是跟踪

$$e_{st} = \lim_{k \rightarrow \infty} (r(k) - y(k)) = \lim_{z \rightarrow 1} ((1 - z^{-1})R(z)[1 - G_{cl}(z)])$$

稳态驱动误差ess
steady-state error
R-B是驱动

$$e_{ss} = \lim_{z \rightarrow 1} \left[(1 - z^{-1}) \frac{1}{1 + GH(z)} R(z) \right]$$

$$GH(z) = GzAS(z)$$

反馈传函为1的话，两者一样，才能共用公式

静态位置误差常数Kp：单位阶跃输入

$$e_{ss} = \frac{1}{1 + K_p}$$

静态速度误差常数 Kv：单位斜坡输入

$$e_{ss} = \frac{1}{K_v}$$

静态加速度误差常数Ka：单位加速度输入t^2/2

$$e_{ss} = \frac{1}{K_a}$$

正向差分方法

1

1

$$s = \frac{1 - z^{-1}}{Tz^{-1}} = \frac{z - 1}{T}$$

反向差分方法

1

1

$$s = \frac{1 - z^{-1}}{T} = \frac{z - 1}{Tz}$$

双线性变换方法

$s \rightarrow z$ 双线性

$$s = \frac{2}{T} \frac{z - 1}{z + 1}$$

$w \rightarrow z$ 双线性

$$w = \frac{2}{T} \frac{z - 1}{z + 1}$$

零极点匹配

2

1

1. 连续控制器

连续控制器：二阶函数
second-order analog filter with

1

1

极点根

阻尼频率

1

2. 离散控制器

欧拉公式

3. 低通匹配

离散1=连续0

PID控制器

P控制器

PD控制器

PI控制器

$$G_a(s) = K \frac{\prod_{i=1}^m (s - a_i)}{\prod_{i=1}^n (s - b_i)}$$

$$G_a(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$
$$s_{1,2} = -\zeta\omega_n \pm j\omega_d$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$G(z) = \alpha \frac{(z + 1)^{n-m-1} \prod_{i=1}^m (z - e^{a_i T})}{\prod_{i=1}^n (z - e^{b_i T})}$$

$$\frac{e^{ix} + e^{-ix}}{2} = \cos x$$

$$\frac{e^{ix} - e^{-ix}}{2i} = \sin x$$

$$e^{ix} = \cos x + i \sin x$$

$$G(1) = G_a(0)$$

$$G(z) = K_p + \frac{K_I}{1 - z^{-1}} + K_D(1 - z^{-1})$$

$$G_c(s) = K_p + K_D s + \frac{K_I}{s}$$

$$G_c(s) = K_p$$

$$G_c(s) = K_p + sK_D$$

$$G_c(s) = \frac{U(s)}{E(s)} = K_p + \frac{K_I}{s}$$

可以用 $1 - z^{-1}$ 来近似 s

$G(s)$ 到 $GD(z)$ 的映射方法

1. 通过性能指标得到期望的 $G_{cl}(s)$
2. 实际的 $G_{cl}(s)$ 闭环传函
3. 通过设计控制器，系数匹配，得到控制器
4. 测试闭环传递函数的系统：频率响应（幅值）

$$H(s) = \frac{Y(s)}{R(s)} = \frac{G(s)C(s)}{1+G(s)C(s)}$$

$$G_{cl}(z) = \frac{C(z)G_{ZAS}(z)}{1 + C(z)G_{ZAS}(z)}$$

频率 得到系统增益从 0 dB（或低频段的最大增益）下降到 -3 dB 的频率

$$\text{带宽} = \frac{J}{2\pi}$$

采样时间

$$T = \frac{1}{30 \times f}$$

- 6.通过s变z的双线新变换得到C(z)

超前补偿器

1. $G_{ZAS}(z)$

$$G_{ZAS}(z) = (1 - z^{-1}) \mathcal{Z} \left\{ \frac{G(s)}{s} \right\}$$

2. z 变 w 的双线性变换

3. $G_{AS}(w)$

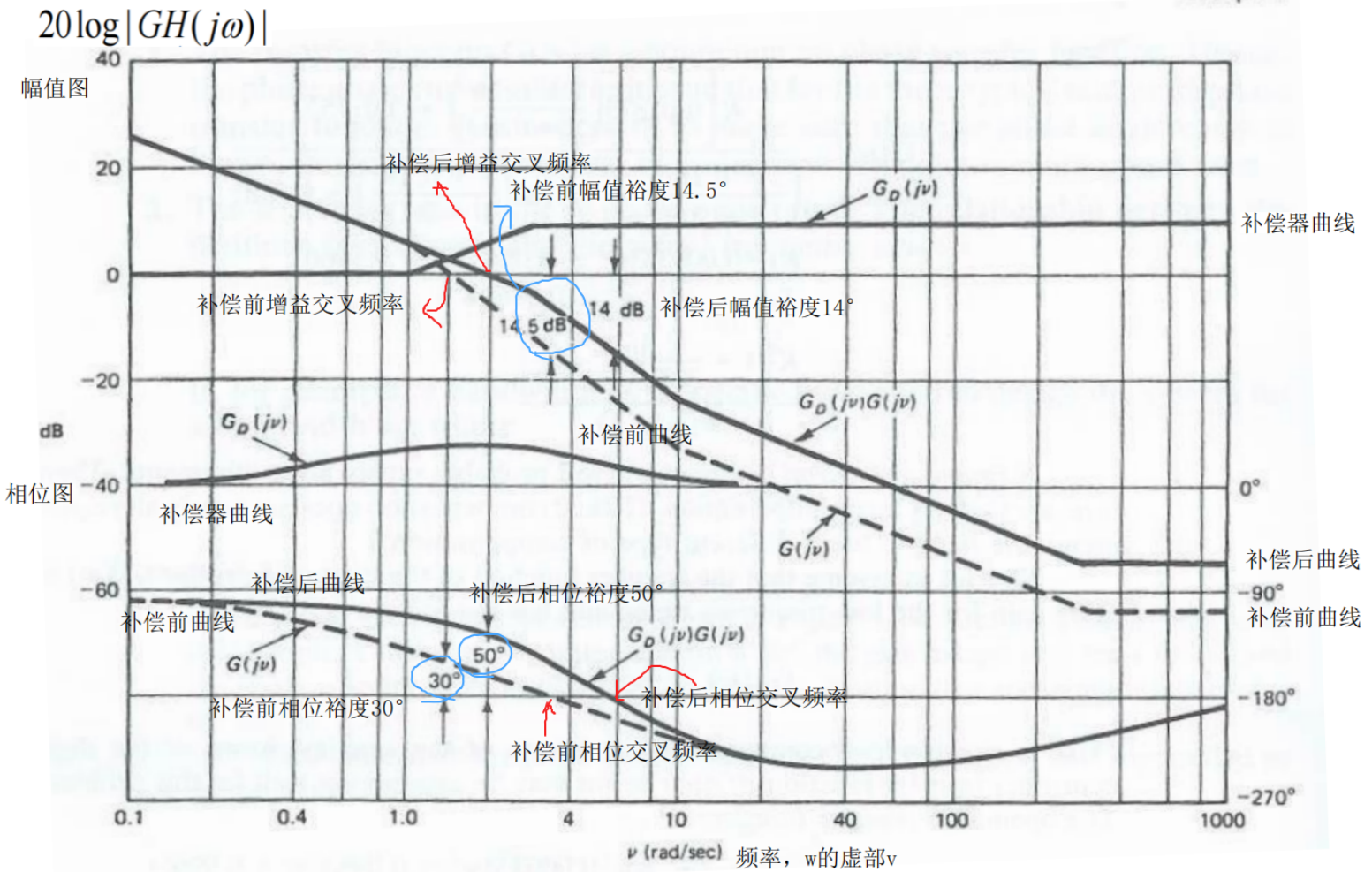
$$G_D(w) = \frac{T_w + 1}{\alpha T_w + 1}, \quad 0 < \alpha < 1$$

4. 相位超前补偿器
phase lead compensator

5. 静速度误差常数

$$K_v = \lim_{w \rightarrow 0} \{ w [G_D(w) G_{ZAS}(w)] \} = K$$

6. 波德图Bode



7. 计算 所需相位裕度

相位裕度差值+8

$$\phi = PM - \phi_0 + (5^0 \text{ to } 10^0)$$

Desired PM

PM of KG(s)

Added to compensate the difference between ϕ_0 and ϕ'_0

一般选择8°

8. 计算 最大相位 对应 的 α

$$\alpha = \frac{1 - \sin \phi}{1 + \sin \phi}$$

9. 解方程：对应最大相位，频率

$$20 \log |G_{ZAS}(jv'_g)| = 10 \log \alpha$$

10. 解出：时间常数T

$$v'_g = \frac{1}{\sqrt{\alpha}T}$$

11. 得到补偿器C(w)

$$G_D(w) = \frac{T\omega + 1}{\alpha T\omega + 1}$$

12. w变为z：得到C(z)

$$G_D(z) = G_D(w) \left|_{w = \frac{2}{T} \frac{z-1}{z+1}}\right.$$

闭环传递函数 $G_{cl}(z)$

$$G_{cl}(z) = \frac{C(z)G_{ZAS}(z)}{1 + C(z)G_{ZAS}(z)}$$

控制器传递函数 $C(z)$

$$C(z) = \frac{1}{G_{ZAS}(z)} \frac{G_{cl}(z)}{1 - G_{cl}(z)}$$

阶跃输入的稳态误差为零

$$G_{cl}(1) = 1$$

死亡节拍，对应的 $G_{cl}(z)$

$$G_{cl}(z) = z^{-k}$$

带入可求 死亡节拍控制器

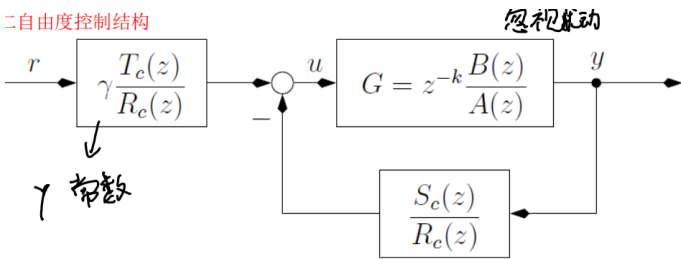
where $k \geq$ the intrinsic delay + relative degree of the discretized process.

无纹波控制器：ripple free

$$U(z) = \frac{Y(z)}{G_{ZAS}(z)} = \frac{Y(z)}{R(z)} \frac{R(z)}{G_{ZAS}(z)} = G_{cl}(z) \frac{R(z)}{G_{ZAS}(z)}$$

1. 写出 $U(z)$ 式子
2. 设计 $G_{cl}(z)=K$ 极点
3. 用 $G_{cl}(1)=1$ 解出 K
4. $C(z)$ 公式

二自由度控制结构



理想的闭环系统输出响应 Y 设计

$$Y_m(z) = \gamma z^{-k} \frac{B_r}{\phi_{cl}} R(z)$$

γ 常数

$$\gamma = \frac{\phi_{cl}(1)}{B_r(1)}$$

输出与输入关系

$$Y = \gamma z^{-k} \frac{T_c B}{A R_c + z^{-k} B S_c} R$$

控制器

$$\frac{B^b T_1}{A^b R_1 + z^{-k} B^b S_1} = \frac{B_r}{\phi_{cl}}$$

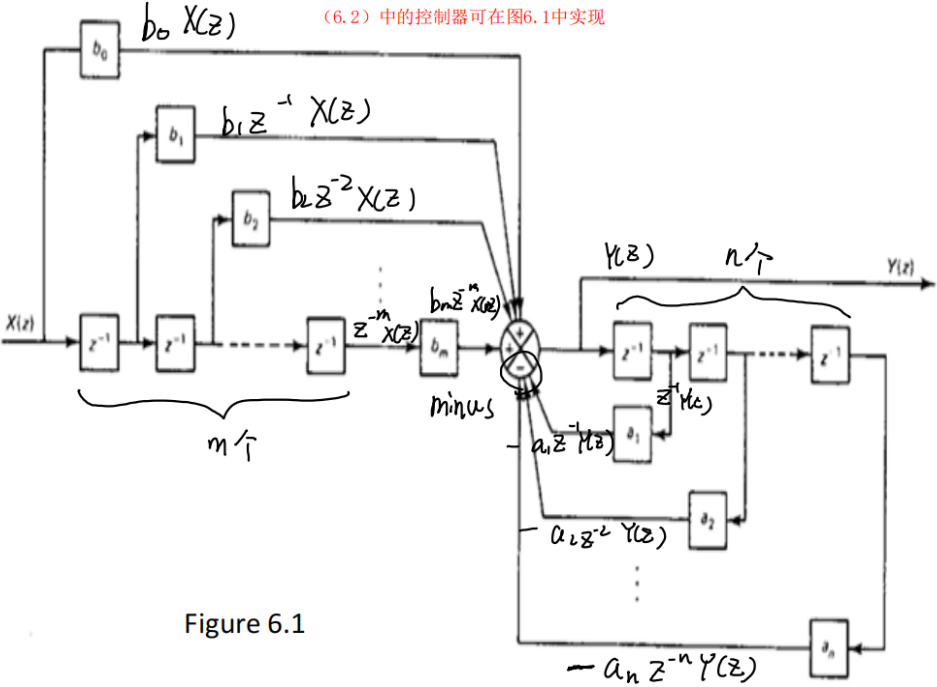
设计 R_1 和 S_1

$$R_c = B^g R_1 \quad S_c = A^g S_1 \quad T_c = A^g T_1$$

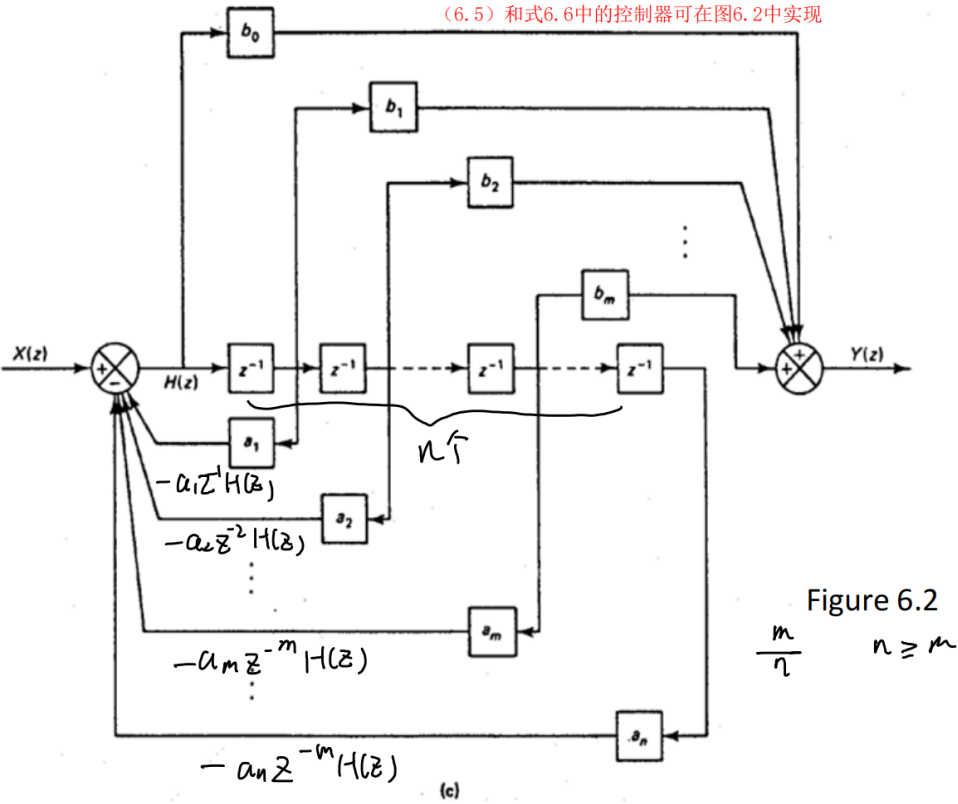
ϕ_{cl}

$$A^b R_1 + z^{-k} B^b S_1 = \phi_{cl}$$

直接编程



标准编程



$$G(z) = G_1(z) G_2(z) \dots G_p(z)$$

$$= \prod_{i=1}^j \frac{1 + b_i z^{-1}}{1 + a_i z^{-1}} \prod_{i=j+1}^p \frac{1 + e_i z^{-1} + f_i z^{-2}}{1 + c_i z^{-1} + d_i z^{-2}}$$

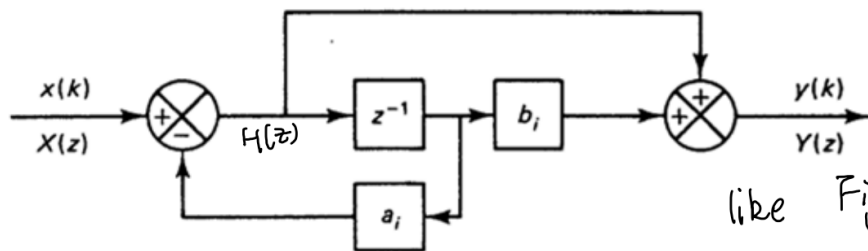
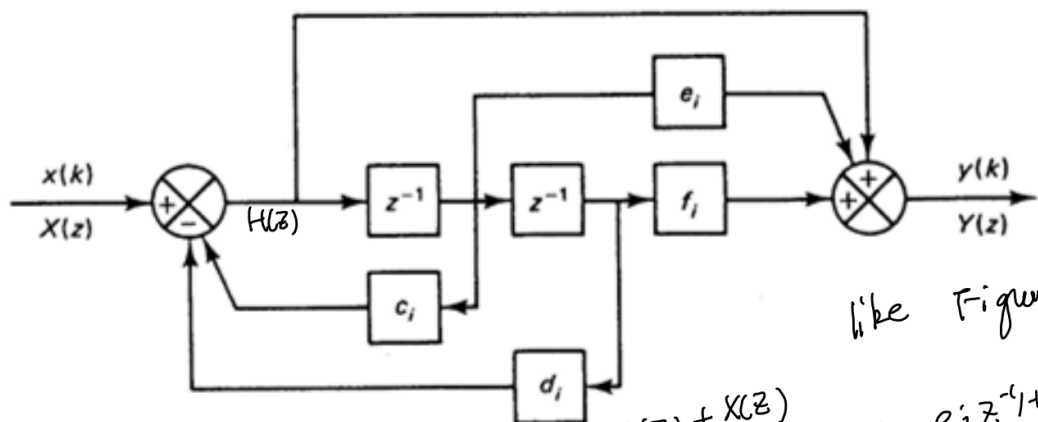


Figure 6.4: $\frac{Y(z)}{X(z)} = \frac{1 + b_i z^{-1}}{1 + a_i z^{-1}}$



$$H(z) = -c_i z^{-1} H(z) - d_i z^{-2} H(z) + X(z)$$

$$Y(z) = H(z) + e_i z^{-1} H(z) + f_i z^{-2} H(z)$$

Figure 6.5: $\frac{Y(z)}{X(z)} = \frac{1 + e_i z^{-1} + f_i z^{-2}}{1 + c_i z^{-1} + d_i z^{-2}} = \frac{Y(z)}{H(z)} \frac{H(z)}{X(z)}$

状态空间

离散时间状态空间模型	2	1	$\mathbf{x}(k+1) = \Phi(T)\mathbf{x}(k) + \Theta(T)\mathbf{u}(k)$ $y(k) = C\mathbf{x}(k)$
1. S域的状态转移矩阵	1	1	$[s\mathbf{I} - \mathbf{A}]^{-1}$
2. t域的状态转移矩阵	1		$\Phi(t) = \mathcal{L}^{-1} \left\{ [s\mathbf{I} - \mathbf{A}]^{-1} \right\}$
3. T域的状态转移矩阵	1		$\Phi(T) = \left[\mathcal{L}^{-1} \left\{ [s\mathbf{I} - \mathbf{A}]^{-1} \right\} \right]_{t=T}$
4. 输入矩阵	1		$\Theta(T) = \int_0^T \Phi(\eta) \, \mathrm{d}\eta \mathbf{B}$

系统传递函数 1

$$\frac{\mathbf{Y}(z)}{\mathbf{U}(z)} = \mathbf{C}[z\mathbf{I} - \mathbf{A}]^{-1}\mathbf{B} + \mathbf{D}$$

$$G(z) = \frac{b_{n-1}z^{n-1} + \cdots + b_1z + b_0}{z^n + \cdots + a_1z + a_0}$$

极点多项式 1

$$|z\mathbf{I} - \mathbf{A}|$$

零点多项式 1

$$\det \begin{bmatrix} z_0\mathbf{I} - \mathbf{A} & -\mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix} = 0$$

相似变换
(similarity transformation) 1 1

$$\mathbf{w}(k) = \mathbf{P}^{-1}\mathbf{x}(k)$$

$$\mathbf{A}_w = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}; \quad \mathbf{B}_w = \mathbf{P}^{-1}\mathbf{B}$$

$$\mathbf{C}_w = \mathbf{C}\mathbf{P}; \quad d_w = d$$

状态空间表示

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}u(k)$$

$$y(k) = \mathbf{C}\mathbf{x}(k) + du(k)$$

特征多项式

$$|\lambda\mathbf{I} - \mathbf{A}| = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$$

Wc 可控矩阵

$$\mathbf{W}_C = \begin{bmatrix} \mathbf{B} & \mathbf{A}\mathbf{B} & \mathbf{A}^2\mathbf{B} & \cdots & \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix}$$

Wo 可观测矩阵

$$\mathbf{W}_O = \begin{bmatrix} \mathbf{C} \\ \mathbf{C}\mathbf{A} \\ \vdots \\ \mathbf{C}\mathbf{A}^{n-1} \end{bmatrix}$$

CCF

Ac

$$\mathbf{A}_C = \begin{bmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & -a_3 & \cdots & -a_{n-1} \end{bmatrix}$$

Bc

$$\mathbf{B}_C = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

~Wc

$$\tilde{\mathbf{W}}_C = \begin{bmatrix} \mathbf{B}_C & \mathbf{A}_C \mathbf{B}_C & \mathbf{A}_C^2 \mathbf{B}_C & \cdots & \mathbf{A}_C^{n-1} \mathbf{B}_C \end{bmatrix}$$

P
原配在前，小三逆转

$$\mathbf{P} = \mathbf{W}_C \tilde{\mathbf{W}}_C^{-1}$$

Cc

$$\mathbf{C}_C = \mathbf{C} \mathbf{P} \quad \mathbf{C}_C = \begin{bmatrix} b_0 & b_1 & \cdots & b_{n-2} & b_{n-1} \end{bmatrix}$$

~Wo

$$\mathbf{W}_O = \begin{bmatrix} \mathbf{C} \\ \mathbf{C} \mathbf{A} \\ \vdots \\ \mathbf{C} \mathbf{A}^{n-1} \end{bmatrix}$$

G(z) 传递函数

$$G(z) = \frac{b_{n-1} z^{n-1} + \cdots + b_1 z + b_0}{z^n + \cdots + a_1 z + a_0}$$

Ao

Co

Bo

Q

~Wo

G(z) 传递函数

$$G(z) = \frac{b_{n-1}z^{n-1} + \dots + b_1z + b_0}{z^n + \dots + a_1z + a_0}$$

法1：匹配求K

$$\alpha(z) = \det[z\mathbf{I} - \mathbf{A} + \mathbf{BK}]$$
$$= z^n + \hat{\beta}_1 z^{n-1} + \dots + \hat{\beta}_{n-1} z + \hat{\beta}_n$$

$$\alpha_C(z) = (z - p_1)(z - p_2) \cdots (z - p_n)$$
$$= z^n + \beta_1 z^{n-1} + \dots + \beta_{n-1} z + \beta_n$$

—————

法2：阿克曼的公式：求K

$$\mathbf{K} = [0 \quad \cdots \quad 0 \quad 1] \mathbf{W}_C^{-1} \alpha_C(\mathbf{A})$$
$$\alpha_C(\mathbf{A}) = \mathbf{A}^n + \beta_1 \mathbf{A}^{n-1} + \dots + \beta_{n-1} \mathbf{A} + \beta_n \mathbf{I}$$

死亡节拍

方法1
闭环系统的特征多项式

方法2
阿克曼公式

$$\alpha_C(z) = z^n = \det [z\mathbf{I} - \mathbf{A} + \mathbf{BK}]$$
$$\mathbf{K} = [0 \quad \cdots \quad 0 \quad 1] \mathbf{W}_C^{-1} \alpha_C(\mathbf{A})$$

Z平面极点转幅度角度

1 1 a b r θ

$$0.888 \pm j0.173 = 0.9054 \pm 0.193$$

$$r = \sqrt{a^2 + b^2}$$
$$\theta = \tan^{-1}(b/a)$$

阻尼比

1 1

$$\zeta = - \frac{\ln r}{\sqrt{(\ln^2 r + \theta^2)}}$$

Z平面转S平面

1 1

$$z = re^{j \theta}$$
$$s = \ln z / T$$

$$\begin{aligned}\bar{\mathbf{x}}(k+1) &= \mathbf{A}\bar{\mathbf{x}}(k) + \mathbf{B}u(k) + \mathbf{L}_o(y(k) - \mathbf{C}\bar{\mathbf{x}}(k)) \\ &= [\mathbf{A} - \mathbf{L}_o\mathbf{C}]\bar{\mathbf{x}}(k) + \mathbf{B}u(k) + \mathbf{L}_oy(k)\end{aligned}$$

Lo 待确定的观测器增益

$$\mathbf{L}_o = \begin{bmatrix} l_1 & l_2 & \cdots & l_n \end{bmatrix}^T$$

方法1：特征方程匹配理想极点

$$|z\mathbf{I} - \mathbf{A} + \mathbf{L}_o\mathbf{C}|$$

方法2：阿克曼直接求

$$\mathbf{L}_o = \alpha_o(\mathbf{A})[\mathbf{W}_o]^{-1} \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

估计误差11

Dr Lee

$$\begin{aligned}\mathbf{x}_e(k) &= \mathbf{x}(k) - \bar{\mathbf{x}}(k) \\ \mathbf{x}_e(k+1) &= [\mathbf{A} - \mathbf{L}_o\mathbf{C}]\mathbf{x}_e(k)\end{aligned}$$

矩阵稳定

1. 该矩阵的**特征值**全部位于复平面的**单位圆**内
2. 当 $n \rightarrow \infty$ 时，矩阵 $\mathbf{A}^n$ 会趋于**零矩阵**

降阶观测器
Lr：计算公式1

$$\begin{aligned}\Rightarrow \bar{\mathbf{x}}_b(k+1) &= [\mathbf{A}_{bb} - \mathbf{L}_r\mathbf{A}_{ab}]\bar{\mathbf{x}}_b(k) + \mathbf{L}_r\mathbf{y}(k+1) + [\mathbf{A}_{ba} - \mathbf{L}_r\mathbf{A}_{aa}]\mathbf{y}(k) + [\mathbf{B}_b - \mathbf{L}_r\mathbf{B}_a]\mathbf{u}(k) \quad \dots(17.4) \\ \mathbf{x}_{b_e}(k+1) &= [\mathbf{A}_{bb} - \mathbf{L}_r\mathbf{A}_{ab}]\mathbf{x}_{b_e}(k) \\ \det[z\mathbf{I} - \mathbf{A}_{bb} + \mathbf{L}_r\mathbf{A}_{ab}] &= \alpha_o(z)\end{aligned}$$

Lr：计算公式21

$$\mathbf{L}_r = \alpha_o(\mathbf{A}_{bb}) \begin{bmatrix} \mathbf{A}_{ab} \\ \mathbf{A}_{ab}\mathbf{A}_{bb} \\ \vdots \\ \mathbf{A}_{ab}\mathbf{A}_{bb}^{n-2} \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

16.2. 验证无差拍错误响应11

$$\mathbf{x}(0) = \begin{bmatrix} a_1 \\ b_1 \end{bmatrix}; \bar{\mathbf{x}}(0) = \begin{bmatrix} a_2 \\ b_2 \end{bmatrix}$$

$$\Rightarrow \mathbf{x}_e(0) = \mathbf{x}(0) - \bar{\mathbf{x}}(0) = \begin{bmatrix} a_1 - a_2 \\ b_1 - b_2 \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\mathbf{x}_e(k+1) = [\mathbf{A} - \mathbf{L}_o \mathbf{C}] \mathbf{x}_e(k)$$

$$\mathbf{x}_e(1) = \begin{bmatrix} -1 & T \\ -1/T & 1 \end{bmatrix} \mathbf{x}_e(0) = \begin{bmatrix} -a + bT \\ -a/T + b \end{bmatrix}$$

$$\mathbf{x}_e(2) = \begin{bmatrix} -1 & T \\ -1/T & 1 \end{bmatrix} \mathbf{x}_e(1) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

现任观察器

$$\begin{aligned}\hat{\mathbf{x}}(k+1) &= \mathbf{A}\bar{\mathbf{x}}(k) + \mathbf{B}u(k) \\ \bar{\mathbf{x}}(k+1) &= \hat{\mathbf{x}}(k+1) + \mathbf{L}_C \left(y(k+1) - \mathbf{C}\hat{\mathbf{x}}(k+1) \right) \\ \bar{\mathbf{x}}(k+1) &= [\mathbf{A} - \mathbf{L}_C \mathbf{C} \mathbf{A}] \bar{\mathbf{x}}(k) + [\mathbf{B} - \mathbf{L}_C \mathbf{C} \mathbf{B}] u(k) \\ &\quad + \mathbf{L}_C y(k+1) \qquad \dots(18.3)\end{aligned}$$

估计器增益Lc

$$\mathbf{L}_C = \alpha_o(\mathbf{A}) \begin{bmatrix} \mathbf{C} \mathbf{A} \\ \mathbf{C} \mathbf{A}^2 \\ \vdots \\ \mathbf{C} \mathbf{A}^n \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

控制律与观测器的组合

控制器极点和估计极点的组合

$$p_1, p_2, \cdots, p_n \quad \cup \quad p_1^O, p_2^O, \cdots, p_n^O$$

特征多项式

$$\alpha_C(z) \alpha_O(z)$$

组合控制器和当前估计器的传递函数

$$\frac{U(z)}{Y(z)} = -z\mathbf{K} \left[z\mathbf{I} - \mathbf{A} + \mathbf{L}_C \mathbf{C} \mathbf{A} + \mathbf{B} \mathbf{K} - \mathbf{L}_C \mathbf{C} \mathbf{B} \mathbf{K} \right]^{-1} \mathbf{L}_C$$

带输入的系统

2

$$u(k) = -\mathbf{K}\mathbf{x}(k) + k_r r(k)$$

K的设计照常进行

传递函数

$$\frac{Y(z)}{R(z)} = k_r \mathbf{C} \left[z\mathbf{I} - \mathbf{A} + \mathbf{B} \mathbf{K} \right]^{-1} \mathbf{B}$$

最优控制：有限时间

1

1

J_i :性能指标或代价函数

表示从初始时间 i 到终止时间 N 的总代价

$\mathbf{x}(N)$: 终止时刻 N 的状态向量

$$J_i = \frac{1}{2} \mathbf{x}^T(N) \mathbf{S}(N) \mathbf{x}(N)$$

$$+ \frac{1}{2} \sum_{k=i}^{N-1} \left(\mathbf{x}^T(k) \mathbf{Q} \mathbf{x}(k) + \mathbf{u}^T(k) \mathbf{R} \mathbf{u}(k) \right)$$

$\mathbf{S}(N)$: 终止代价矩阵

3

2

 $\mathbf{S}(N)=0$

$\mathbf{Q}$: 状态代价矩阵

系统在每个时刻偏离目标状态的代价

1

1

1. $k=N-1$ 开始带入求解2. 一直代到 $k=0$

$\mathbf{x}(k)$ 每个时间步的状态

$\mathbf{R}$: 控制代价矩阵,
控制输入的“代价”

$\mathbf{u}(k)$ 控制输入

最优控制：无限时间

$$J = \frac{1}{2} \sum_{k=0}^{\infty} \left(\mathbf{x}^T(k) \mathbf{Q} \mathbf{x}(k) + r u^2(k) \right).$$

反馈增益矩阵

$$\mathbf{K}(k) = \left(\mathbf{B}^T \mathbf{S}(k+1) \mathbf{B} + \mathbf{R} \right)^{-1} \mathbf{B}^T \mathbf{S}(k+1) \mathbf{A}$$

代价函数的权重矩阵

$$\mathbf{S}(k) = \left[\mathbf{A} - \mathbf{B} \mathbf{K}(k) \right]^T \mathbf{S}(k+1) \left[\mathbf{A} - \mathbf{B} \mathbf{K}(k) \right] \\ + \mathbf{K}^T(k) \mathbf{R} \mathbf{K}(k) + \mathbf{Q}$$

$$\mathbf{u}^*(k) = -\mathbf{K}(k) \mathbf{x}(k); J_k^* = \frac{1}{2} \mathbf{x}^T(k) \mathbf{S}(k) \mathbf{x}(k)$$

使 J 最小的控制律
optimal control law

$$\mathbf{u}^*(k) = -\mathbf{K} \mathbf{x}(k),$$

最优控制增益

$$\mathbf{K} = \left(\mathbf{B}^T \mathbf{S} \mathbf{B} + r \right)^{-1} \mathbf{B}^T \mathbf{S} \mathbf{A}$$

$S > 0$ 解下式

$$\mathbf{S} = \mathbf{A}^T \mathbf{S} \mathbf{A} + \mathbf{Q} - \mathbf{A}^T \mathbf{S} \mathbf{B} \left(r + \mathbf{B}^T \mathbf{S} \mathbf{B} \right)^{-1} \mathbf{B}^T \mathbf{S} \mathbf{A}.$$

1. 解 $\mathbf{S}$ 2. 解 $\mathbf{K}$ 3. 带入 $\mathbf{u}^*$

resulting closed-loop state equation

4. 带入得到 $\mathbf{x}(k+1)$